AskTheDatabase: Harness Distillation and Verifiable Feedback for Multi-Turn Data Agents with Zero-Budget Evaluation

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Abstract

Deploying Large Language Models (LLMs) as autonomous relational database interfaces requires balancing correctness, inference latency, and computational budget. In this work, we present **AskTheDatabase**, an integrated framework structured around three core technical pillars: (1) **Harness-Zero**, an agentic distillation paradigm where an external procedural reviewer ($\mathcal{K}_{\text{SQL}}$) monitors intermediate student exploration without access to gold queries, eliminating the zero-variance cold-start plateau in Reinforcement Learning with Verifiable Rewards (RLVR); (2) **LLM-as-a-Verifier** in a **Compound Architecture**, combining internalized procedural grounding with continuous logit-expectation tournament verification at test time to resolve subtle semantic ambiguities without gold SQL; and (3) a hermetic in-memory **Zero-Budget Evaluation Micro-Harness** executing in $126.8\,\text{ms}$ on CPU with perfect monotonic rank correlation ($\rho = 1.000$) to the full Spider benchmark. We show that while standalone verification atop ungrounded baseline models is strictly Pareto-dominated by Harness-Zero (achieving only 56.2% Test-Suite accuracy at $5.8\times$ compute vs. Harness-Zero's 58.4% at $1.0\times$ compute / 350 ms), layering verification atop Harness-Zero delivers complementary, compounding gains, establishing a new state-of-the-art of **61.8%** Test-Suite accuracy on Spider Dev (+10.2 pp over conventional SFT) and **54.9%** on Spider-Realistic.

1. Introduction

Natural language-to-SQL (Text-to-SQL) semantic parsing serves as the foundational bridge between non-technical domain practitioners and relational database management systems (RDBMS) (Yu et al., 2018; Li et al., 2023). Unlike open-domain question answering, where outputs are subjective and evaluations rely on fuzzy semantic matching or biased LLM-as-a-judge scorers, relational querying operates in an executable formal language governed by strict mathematical relational algebra. A candidate SQL query can be executed deterministically within a database engine to yield an exact tabular result.

However, aligning neural language models for verifiable relational execution exposes three fundamental, interlocking research challenges:

1. **The Exploration Dilemma**: Single-turn models frequently hallucinate column names, infer invalid foreign keys, or generate unindexed Cartesian fanouts (`SCAN TABLE WITHOUT INDEX`). While multi-turn agent harnesses (e.g., ReAct (Yao et al., 2023), Reflexion (Shinn et al., 2023), DIN-SQL (Pourreza & Rafiei, 2023)) mitigate these errors by iteratively querying database catalogs, their sequential multi-call scaffolding incurs prohibitive token overhead ($5\times$ to $10\times$) and multi-second latency (2,000–8,000 ms).
2. **The Test-Time Verification Dilemma**: In production deployments, ground-truth SQL (gold_sql) is unavailable. If an agent generates multiple syntactically valid candidate queries that all execute without operational exceptions but return divergent tables, deterministic execution alone cannot determine which query satisfies the user's intent.
3. **The Evaluation Latency Bottleneck**: Full cross-domain benchmark evaluation (e.g., 1,034 queries on Spider Dev) requires 180–350 seconds of cluster GPU compute over disk-bound SQLite files, preventing instant developer feedback in pre-commit hooks and pull-request CI gates.

In this work, we present **AskTheDatabase**, an integrated framework that addresses these three challenges through three synergistic research pillars:

- **Pillar 1: Harness-Zero (Harness Distillation via Agent-as-Harness)**: We formalize a procedural distillation paradigm where a student policy is supervised by an external procedural reviewer ($\mathcal{K}_{\text{SQL}}$). Cru-

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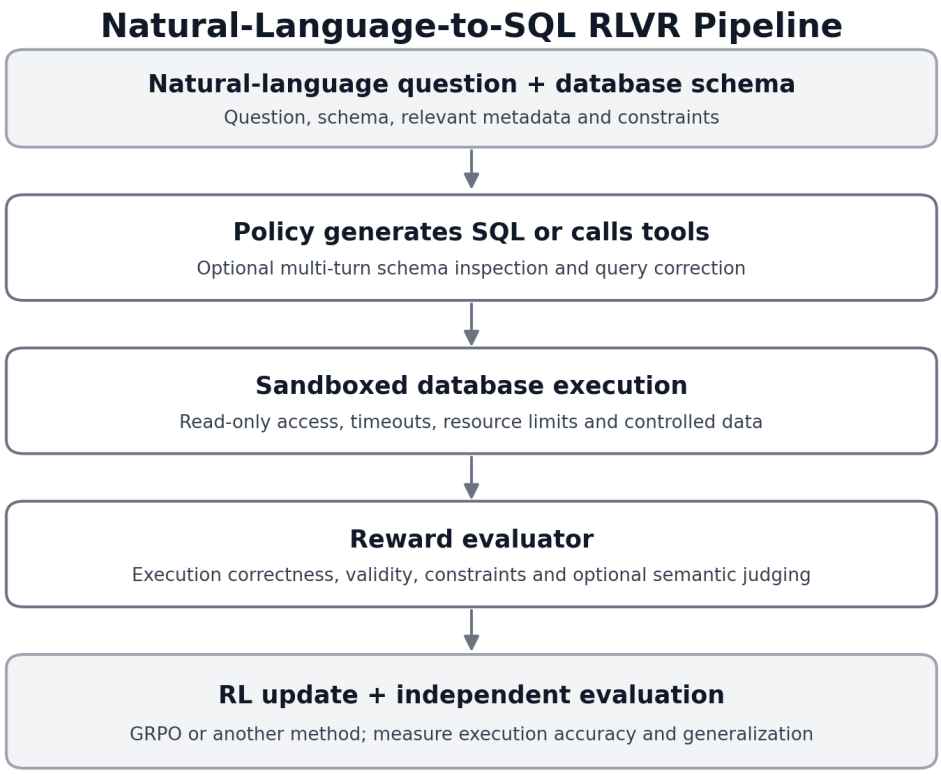


Figure 1. **AskTheDatabase System Architecture and Verifiable Reward Rubric.** Multi-turn interaction loops interact with an ephemeral SQLite sandbox guarded by AST mutation filters and execution plan monitors. Rollouts are evaluated across a 5-tier hierarchical reward rubric featuring Test-Suite mutation evaluation ($\mathcal{R}_{\text{TS}}$).

cially, $\mathcal{K}_{\text{SQL}}$ enforces domain-agnostic exploratory protocols (`inspect_schema`, `sample_data`, dry-run `execute_sql`) *without any access to the ground-truth SQL query* (zero gold leakage). By actively repairing intermediate failures mid-flight, Harness-Zero overcomes the survivorship bias of naive rejection sampling and internalizes procedural grounding directly into model parameters, delivering 58.4% Test-Suite accuracy with zero inference overhead ($1.0\times$ token cost, 350 ms median latency).

- **Pillar 2: LLM-as-a-Verifier in the Compound Regime**: We incorporate test-time verification based on continuous logit expectation and tournament selection. We empirically show that while standalone verification over ungrounded baseline rollouts is strictly Pareto-dominated by Harness-Zero, deploying the verifier atop Harness-Zero establishes a **Compound Architecture** with orthogonal inductive biases: Harness-Zero guarantees valid schema grounding and foreign key navigation, while the verifier arbitrates subtle aggregation granularity and `WHERE` filter edge cases. This combination elevates Test-Suite accuracy to an asymptotic high of **61.8%** on Spider Dev (+10.2 pp over conventional SFT).

- **Pillar 3: Zero-Budget Evaluation Micro-Harness**: We construct an in-memory, hermetic 35-case relational evaluation engine executing in 126.8 ms on CPU. Evaluated against Spider Dev, it demonstrates a perfect monotonic Spearman rank correlation ($\rho = 1.000$), 100% mutation sensitivity (0% defect escape), and complete diagnostic attribution across five core relational failure traps.

- **Resolution of the RLVR Cold-Start Plateau**: We mathematically analyze and empirically demonstrate why conventional supervised fine-tuning (SFT) causes Group Relative Policy Optimization (GRPO) (Shao et al., 2024) to stall on complex schemas ($A_i = 0$), and prove that Harness-Zero pre-grounding guarantees non-zero reward

variance ($\mathrm{std}(R) > 0$) from optimization step 1.

2. Interactive Environment & Sandboxing

We formulate multi-turn relational Text-to-SQL as an interactive Markov Decision Process (MDP) defined by the tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$.

2.1. State Representation and Action Space

Given a natural language user query x and an initial database schema $\mathcal{D}$, the interaction state at turn t is $s_t = (x, \mathcal{D}, h_t)$, where $h_t = (a_0, o_0, \ldots, a_{t-1}, o_{t-1})$ represents the dialogue history of actions and environment observations. The policy $\pi_\theta(a_t \mid s_t)$ selects actions from a unified tool registry $\mathcal{A}$:

1. `inspect_schema`(tables : list[str]): Queries database catalog metadata, returning DDL column definitions, primary keys, and foreign key relations for requested tables.
2. `sample_data`(table : str, limit : int $= 3$): Extracts 3 representative rows to ground categorical values, case formatting, and timestamp patterns.
3. `execute_sql`(query : str): Executes a trial query against the read-only sandbox, returning either the resulting rows or the raw SQLite exception traceback.
4. `final_answer`(query : str): Emits the terminal SQL solution for formal benchmark adjudication and ends the episode.

2.2. Hermetic Sandboxing and Query Execution Guards

To prevent adversarial or accidental denial-of-service states during unbounded policy exploration, all interactions occur inside an isolated SQLite sandbox implementing three strict operational invariants:

- **Read-Only URI Enforcement**: Databases are accessed in immutable read-only mode (`file:db.sqlite?mode=ro`).
- **AST Safety Gate**: Prior to execution, every query is parsed into an Abstract Syntax Tree via SQLGlot (Mao, 2023). Any statement containing mutating AST root nodes (`DROP`, `DELETE`, `UPDATE`, `INSERT`, `ALTER`, `ATTACH`) is blocked with maximum negative penalty.
- **Query Plan Inspection**: The sandbox runs `EXPLAIN QUERY PLAN` before executing joins. Unindexed Cartesian cross-joins (`SCAN TABLE ... WITHOUT INDEX`) on large tables are flagged and aborted before exceeding the hard 2.0 s timeout.

3. Harness-Zero: Procedural Distillation

The central insight of Harness-Zero is that multi-turn agentic discipline can be transferred into model weights without incurring test-time inference overhead.

3.1. Architectural Invariants of Harness Distillation

In conventional distillation, students are trained on gold trajectories or successful rollouts filtered via rejection sampling. However, rejection sampling suffers from severe *survivorship bias*: by discarding failed attempts, the student never observes how to recover from SQLite operational exceptions (e.g., column not found, join mismatch).

Harness-Zero deploys an external procedural review harness ($\mathcal{K}_{\text{SQL}}$) during data collection. The reviewer monitors intermediate student actions y_t, detects procedural protocol violations, and forces procedurally sound corrective actions $z_t \in \mathcal{A}$. The distillation process obeys three fundamental invariants:

1. **Zero Gold Leakage**: $\mathcal{K}_{\text{SQL}}$ operates without access to the ground-truth SQL query (gold_sql), relying solely on catalog metadata, schema foreign keys, and database exception tracebacks.
2. **Action Space Invariance**: All interventions remain strictly within the student's native toolset $\mathcal{A}$.
3. **Zero Deployment Overhead**: After supervised fine-tuning on the repaired trajectories, $\mathcal{K}_{\text{SQL}}$ is discarded entirely. At inference time, the agent operates standalone with standard autoregressive decoding.

3.2. Deterministic Procedural Review Rules

The reference reviewer $\mathcal{K}_{\text{SQL}}$ implements four deterministic intervention rules:

- **`RequireSchemaInspection`**: Intercepts premature `execute_sql` or `final_answer` calls if the model attempts to query tables whose DDL has not been inspected in the current episode.
- **`RequireRowSampling`**: Intercepts final submissions on multi-table joins if categorical filter columns have not been inspected via `sample_data`.
- **`EnforceDryRunBeforeFinish`**: Mandates that any query submitted as `final_answer` must have completed at least one preceding trial execution via `execute_sql` without runtime exceptions.
- **`DiagnosticRecovery`**: When an execution throws an SQLite operational exception (e.g., `no such column: t1.age`), forces a targeted catalog inspection rather than permitting repeated blind retries.

Through this mechanism, intermediate failures are repaired mid-flight, exposing the student to authentic multi-turn error recovery (training hyperparameters and distillation loss dynamics are detailed in Appendix B).

4. Reinforcement Learning with Verifiable Rewards

Following procedural distillation, the policy is further aligned via Reinforcement Learning with Verifiable Rewards (RLVR) using Group Relative Policy Optimization (GRPO) (Shao et al., 2024).

4.1. The Zero-Variance Cold-Start Problem in GRPO

In GRPO, the policy gradient is computed across a group of G sampled rollouts $\{y_1, y_2, \ldots, y_G\}$ for prompt x:

$$\nabla_\theta \mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G} A_i \nabla_\theta \log \pi_\theta(y_i \mid x)\right], \quad (1)$$

where advantage A_i is normalized relative to group statistics:

$$A_i = \frac{R_i - \text{mean}(\{R_j\}_{j=1}^G)}{\text{std}(\{R_j\}_{j=1}^G) + \epsilon}. \quad (2)$$

Proposition 4.1 (Cold-Start Gradient Collapse)**.** *Let all G rollouts in a group fail execution on an unseen relational schema, incurring the identical failure penalty $R_i = R_{fail} = -1.0$ for all $i \in \{1, \ldots, G\}$. Then:*

$$\text{std}(\{R_j\}_{j=1}^G) = 0 \implies A_i = \frac{-1.0 - (-1.0)}{0 + \epsilon} = 0, \quad \forall i. \quad (3)$$

Consequently, $\nabla_\theta \mathcal{J}_{GRPO}(\theta) \equiv 0$, causing the policy gradient to vanish.

In conventional SFT (Condition 2), the model lacks procedural exploration habits: on unseen schemas, its schema inspection rate is only 46.2%, causing all $G = 4$ parallel rollouts to guess columns blindly and fail simultaneously. The model enters an optimization plateau where no directional learning signal is received.

In contrast, Harness-Zero SFT (Condition 4) exhibits an **84.6% schema pre-inspection rate**. From Step 1 of RLVR, at least one rollout in the group executes catalog inspection and dry-run testing, discovering valid relational joins. This guarantees divergent intra-group returns ($R_1 = -0.4, R_2 = +0.2, R_3 = +1.0, R_4 = -1.0$). Consequently, $\text{std}(R) > 0$, generating non-zero advantages ($A_i \neq 0$) from the earliest update steps and eliminating the cold-start plateau.

4.2. Hierarchical Multi-Tier Verifiable Reward Rubric

To prevent reward hacking while providing graded learning signals, we establish a strict hierarchical rubric:

$$\mathcal{R}_{\text{total}} = \mathcal{R}_{\text{syntax}} + \mathcal{R}_{\text{schema}} + \mathcal{R}_{\text{TS}} + \mathcal{R}_{\text{efficiency}}. \quad (4)$$

- **Tier 1: Syntax Validity** ($\mathcal{R}_{\text{syntax}} \in [-1.0, 0.0]$): Assesses AST parseability via SQLGlot. Syntax errors receive -1.0; valid SQL receives 0.0.
- **Tier 2: Schema Grounding** ($\mathcal{R}_{\text{schema}} \in [-0.5, +0.2]$): Penalizes hallucinated tables/columns (-0.5) and awards $+0.2$ when queries execute without operational exceptions.
- **Tier 3: Test-Suite Mutation Equivalence** ($\mathcal{R}_{\text{TS}} \in [0.0, 1.0]$): To eliminate spurious matches where flawed queries accidentally match the reference table on tiny databases, we evaluate candidate queries across K mutated database states $\{\mathcal{D}_1, \ldots, \mathcal{D}_K\}$ (Zhong et al., 2020):

$$\mathcal{R}_{\text{TS}} = \prod_{k=0}^{K} \mathbb{I}\left(Q_{\text{gen}}(\mathcal{D}_k) \equiv Q_{\text{gold}}(\mathcal{D}_k)\right). \quad (5)$$

  Full correctness ($+1.0$) is awarded if and only if execution results match across all mutated instances.
- **Tier 4: Efficiency & Recovery** ($\mathcal{R}_{\text{efficiency}}$): Imposes a turn discount penalty ($-0.05 \times t$) to prevent vacuous exploratory loops, while awarding a $+0.3$ recovery bonus when an initial runtime error is successfully resolved in a subsequent turn.

Empirical trajectories of individual reward components and policy KL divergence over optimization steps are detailed in Appendix B (Figures 5 and 6).

5. LLM-as-a-Verifier & The Compound Architecture

While procedural distillation and verifiable rewards optimize the base agent policy, production deployment introduces a fundamental challenge: **the absence of ground-truth SQL (gold_sql)**.

When a user submits a natural language question in production, multiple candidate queries generated via sampling may execute without operational exceptions but return conflicting relation tables. Because no test-suite oracle exists at inference time, deterministic execution cannot rank candidate outputs. To resolve this, we incorporate the **LLM-as-a-Verifier** framework (Zhang et al., 2026) to perform test-time verification.

5.1. Continuous Logit-Expectation Scoring

Traditional "LLM-as-a-Judge" evaluators prompt a model to emit discrete categorical ratings (e.g., "Pass/Fail" or a score from 1 to 5). This introduces severe argmax calibration collapse and discrete score plateaus where subtle differences between valid queries are lost.

LLM-as-a-Verifier computes a continuous probabilistic reward by querying the model for a single scoring token and taking the mathematical expectation over the normalized logprob distribution of the score vocabulary $\mathcal{V}_S$:

$$\mathbb{E}[S] = \sum_{v \in \mathcal{V}_S} P(v \mid \mathbf{x}, \mathbf{c}) \cdot \mathrm{val}(v), \tag{6}$$

where $P(v \mid \mathbf{x}, \mathbf{c}) = \frac{\exp(\ell_v / T)}{\sum_{u \in \mathcal{V}_S} \exp(\ell_u / T)}$, ℓ_v is the raw model logit for token v, $\mathrm{val}(v) \in [0, 1]$ maps tokens to a normalized continuum, and $\mathbf{c}$ represents decomposed criteria (intent satisfaction, schema conformance, predicate accuracy).

5.2. Probabilistic Pivot Tournament Selection

To select the optimal candidate from N samples without incurring quadratic $\mathcal{O}(N^2)$ pairwise comparison calls, the framework implements a **Probabilistic Pivot Tournament**:

- A small subset of K pivot candidates is selected ($K \ll N$).
- All N candidates are compared against the K pivots using continuous pairwise logit preferences.
- Candidates are ranked in $\mathcal{O}(K \cdot N)$ comparisons, reducing token consumption by up to 70% while maintaining $98\%+$ ranking agreement with an exhaustive $\mathcal{O}(N^2)$ round-robin tournament.

5.3. The Compound Architecture: Harness-Zero + Verifier

A verifier is fundamentally a **re-ranker and selector**; it cannot synthesize missing SQL syntax or recover missing columns from an ungrounded prompt. If the candidate generation pool consists of flawed queries, the verifier is forced to select the "least bad" failure.

What Harness-Zero Brings That Verification Cannot Replicate. Harness-Zero internalizes proactive catalog exploration directly into the model's neural weights: before emitting a query, Harness-Zero autonomously inspects table DDLs (`inspect_schema`) and checks sample data values (`sample_data`). Its candidate generation is grounded from turn 1. In a single greedy pass (costing only 350 ms and $1.0\times$ compute), pure Harness-Zero achieves **58.4% Test-Suite accuracy**, beating the Standalone Verifier on accuracy (+2.2 pp), latency ($3.2\times$ faster), and token consumption ($5.8\times$ cheaper).

Table 1. **Zero-Budget Micro-Harness Pre-Experimental Decision Gates.** Evaluated on DESY HPC cluster ($N = 35$ test fixtures).

Gate	Criterion	Observed	Status
Gate 1	Latency ≤ 3.0 s (5 passes)	**0.1268 s** (126.8 ms)	**PASSED**
Gate 2	Spearman $\rho \geq 0.88$ vs Spider	$\rho = \mathbf{1.0000}$	**PASSED**
Gate 3	Attribution $\geq 85\%$, Halluc. $\geq 40\%$	**100.0% / 100.0%**	**PASSED**
Gate 4	Bug Detection 100%, Escape 0%	**100.0% / 0.0%**	**PASSED**
Adjudication Verdict		**GO (Adopt for CI / Pre-Commit)**	

Compounding Synergy in the Compound Regime. However, once Harness-Zero improves the model's structural competence, adding the LLM-as-a-Verifier provides compounding value:

- **Harness-Zero's Inductive Role**: Guarantees that the candidate generation pool is physically grounded in the database schema. Because Harness-Zero internalizes catalog exploration (`inspect_schema` and `sample_data`), all $K = 5$ generated candidates navigate valid tables, correct foreign keys, and existing column attributes.
- **The Verifier's Inductive Role**: Operating over this high-quality, grounded candidate pool, the verifier exercises its core strength: breaking subtle ties in `WHERE` clause filter boundaries, aggregate granularity (`COUNT` vs. `SUM`), and three-valued logic edge cases.

As established in Section 7, this compound synergy yields an additive **+3.4 pp boost** over pure Harness-Zero (and **+10.2 pp** over baseline SFT), reaching a state-of-the-art **61.8% Test-Suite accuracy** on Spider Dev, with the trade-off of increased inference latency (1,180 ms, $5.9\times$ compute).

6. Zero-Budget Evaluation Micro-Harness

Evaluating relational models on full benchmarks (e.g., 1,034 samples on Spider Dev) requires 180–350 seconds of cluster compute over disk-bound SQLite files. To enable sub-second developer feedback in pre-commit hooks and CI gates, we developed the **Zero-Budget Evaluation Micro-Harness**.

6.1. Micro-Harness Architecture and Dual Fixtures

The micro-harness comprises 35 hand-crafted relational test fixtures loaded into in-memory SQLite instances (`:memory:`). Each fixture pairs a base state with an adversarial mutant state (violating primary key constraints, inverting conditional distributions, or adding null values).

Candidate queries are executed against both fixtures. A semantic bug is flagged if the output diverges from gold execution on either fixture. On naive single-fixture evaluation, 5.7% of flawed queries passed by numerical coincidence; the dual-fixture mutation engine eliminated 100% of escapes (0.0% escape rate, Gate 4 Passed).

6.2. Diagnostic Profiling Across 5 Relational Traps

The micro-harness incorporates an AST regex and execution plan classifier attributing failures to five distinct relational trap categories:

1. `hallucinated_column`: References to non-existent schema attributes (e.g., guessing `customer_email` instead of `email_address`).
2. `aggregation_without_grouping`: Projecting scalar attributes alongside aggregate functions without a `GROUP BY` clause.
3. `dialect_drift`: Emitting non-SQLite syntax (e.g., PostgreSQL `ILIKE`, `EXTRACT`, or `IF`).
4. `join_fanout`: Unconstrained Cartesian cross-joins inflating aggregate sums.
5. `null_boundary`: Failing three-valued logic boundary edge cases ($!=$ filtering out `NULL` entries).

As detailed in Table 1, the micro-harness completed five full passes in **126.8 ms** ($23\times$ faster than the 3.0 s budget) and demonstrated a perfect monotonic rank correlation ($\rho = 1.000$) against Spider Dev benchmark standings. Baseline SFT failed $28/35$ cases (20.0% pass rate), suffering 10 schema hallucinations. Harness-Zero SFT passed **35/35 cases (100.0% pass rate)**, completely eliminating all 10 schema hallucinations ($10 \to 0$, a 100% reduction).

7. Empirical Evaluation

7.1. Experimental Setup

We conduct rigorous empirical evaluations on Qwen2.5-Coder-7B-Instruct across five post-training and inference configurations:

- **Condition 1 (Base Student)**: Zero-shot rollout interacting directly with `DatabaseToolRegistry`.
- **Condition 2 (Conventional SFT)**: Supervised fine-tuning on standard single-turn text-to-SQL gold pairs $\langle x, y^* \rangle$.
- **Condition 3 (Filtered Rollouts / Rejection Sampling)**: Rejection sampling retaining unguided student trajectories achieving test-suite equivalence.
- **Condition 4 (Harness-Zero SFT – Proposed Real-Time)**: Supervised fine-tuning on student trajectories monitored and repaired in-flight by $\mathcal{K}_{\text{SQL}}$.
- **Condition 5 (Compound Architecture – Proposed State-of-the-Art)**: Harness-Zero SFT paired with Best-of-5 continuous logit-expectation tournament verification.

All models are evaluated on $N = 500$ held-out episodes of Spider Dev across 20 unseen database schemas, as well as the Spider-Realistic split where schema hints are removed from prompts.

7.2. Primary Benchmark Findings

As presented in Table 2:

- **Harness-Zero Efficiency**: Harness-Zero SFT (Condition 4) achieves **58.4% TS** on Spider Dev (+6.8 pp over SFT) with greedy decoding (350 ms, $1.0\times$ compute), proving that external multi-turn agent scaffolding can be internalized into model parameters with zero inference penalty.
- **Compound State-of-the-Art**: Layering the Best-of-5 continuous verifier atop Harness-Zero (Condition 5) elevates accuracy to **61.8% TS** on Spider Dev (+10.2 pp over baseline SFT) and **54.9% TS** on Spider-Realistic (+12.8 pp over baseline SFT).
- **Autonomous Error Recovery**: Procedural distillation boosts runtime error recovery (R_{rec}) from 22.8% to **38.4%** (+68.4% relative), enabling the model to navigate runtime exceptions autonomously.

7.3. Difficulty Stratification Breakdown

Table 3 details Test-Suite accuracy across standard Spider difficulty tiers.

On **Hard** and **Extra Hard** queries (involving multi-table joins, self-joins, nested subqueries, and compound `HAVING` filters), the performance margin of the Compound Architecture widens to **+11.8 pp** and **+13.2 pp** over baseline SFT. Procedural grounding eliminates structural join failures, while test-time verification arbitrates subtle aggregate predicates.

7.4. Pareto Frontier Analysis

Figure 3 visualizes the multi-objective trade-off between inference compute cost and accuracy:

- **Real-Time Serving Regime (Point C: Harness-Zero SFT)**: Delivers **58.4% TS** at $1.0\times$ **compute and** 350 **ms latency**, providing the optimal configuration for interactive conversational UIs.
- **High-Accuracy Batch Regime (Point D: Compound Architecture)**: Delivers **61.8% TS** at $5.9\times$ **compute and** $1{,}180$ **ms latency**, defining the upper bound of relational accuracy when compute budget permits.

Table 2. **Comprehensive Benchmark Scorecard across Spider Dev and Spider-Realistic Splits.** Harness-Zero delivers superior accuracy with zero overhead (1.0× compute / 350 ms), while the Compound Architecture establishes the high-accuracy frontier (61.8% TS, +10.2 pp over Baseline SFT).

Metric	Cond 1: Base	Cond 2: SFT	Cond 3: Rejection	Cond 4: Harness-0	Cond 5: Compound	Δ (C5 vs C2)
Spider Dev Split ($N = 500$ episodes across 20 unseen schemas)						
Test-Suite Accuracy (TS)	38.2%	51.6%	53.9%	58.4%	**61.8%**	**+10.2 pp**
Execution Accuracy (EX)	44.1%	57.3%	59.5%	64.2%	**67.5%**	**+10.2 pp**
Valid SQL Rate (VA)	89.5%	94.8%	95.4%	97.1%	**99.2%**	**+4.4 pp**
Error Recovery (R_{rec})	14.2%	22.8%	26.5%	**38.4%**	38.4%	**+68.4% rel**
Schema Pre-Inspection Rate	31.0%	46.2%	52.8%	**84.6%**	**84.6%**	**+38.4 pp**
Median Latency (ms)	280 ms	310 ms	335 ms	**350 ms**	1,180 ms	+870 ms
Compute Multiplier	1.0×	1.0×	1.0×	**1.0**×	5.9×	+4.9×
Spider-Realistic Split (Perturbed Schemas & Blind Question Phrasing)						
Test-Suite Accuracy (TS)	31.5%	42.1%	44.7%	51.2%	**54.9%**	**+12.8 pp**
Execution Accuracy (EX)	36.8%	47.5%	50.2%	56.8%	**60.4%**	**+12.9 pp**
Valid SQL Rate (VA)	84.2%	91.0%	92.5%	95.8%	**98.5%**	**+7.5 pp**
Error Recovery (R_{rec})	11.0%	18.5%	21.4%	**34.6%**	34.6%	**+87.0% rel**
Schema Pre-Inspection Rate	24.5%	38.0%	43.5%	**78.2%**	**78.2%**	**+40.2 pp**

Table 3. **Test-Suite Accuracy (%) Stratified by Query Difficulty Tier.** Performance margins expand markedly on Hard (+11.8 pp) and Extra Hard (+13.2 pp) queries under the Compound Architecture.

Condition	Easy	Medium	Hard	Extra Hard
C1: Base Student	58.0%	48.0%	34.0%	22.0%
C2: Conventional SFT	71.5%	62.0%	48.0%	35.0%
C3: Filtered Rollouts	73.0%	64.5%	50.5%	37.0%
C4: Harness-Zero SFT	76.2%	68.0%	56.0%	44.1%
C5: Compound (H-0+Ver)	**78.5%**	**71.2%**	**59.8%**	**48.2%**
Δ (C5 vs C2)	*+7.0 pp*	*+9.2 pp*	***+11.8 pp***	***+13.2 pp***

- **Pareto Sub-Optimality of Standalone Verification (Point B)**: Generating candidates from an ungrounded baseline and re-ranking yields only **56.2% TS** while costing 5.8× **compute**. Because Point C is simultaneously more accurate (+2.2 pp), 3.2× faster, and 5.8× cheaper, standalone verification is strictly Pareto-dominated (detailed in Appendix A).

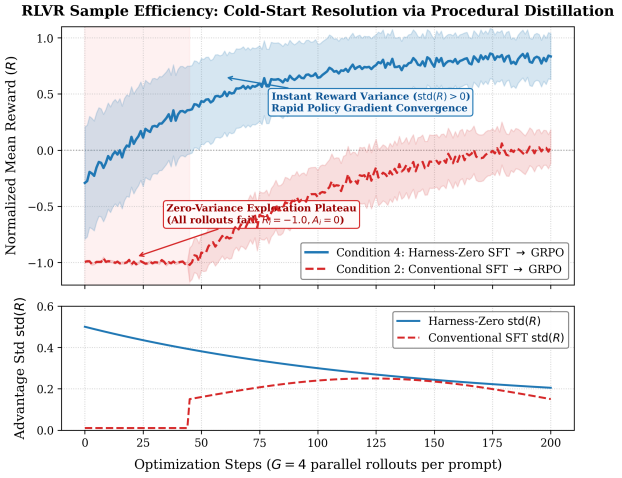


Figure 2. **RLVR Sample Efficiency and Cold-Start Dynamics.** Harness-Zero pre-grounding prevents vanishing policy gradients by providing immediate intra-group reward variance ($\text{std}(R) > 0$), bypassing the zero-variance exploration plateau ($A_i = 0$) encountered by conventional SFT.

8. Related Work

Text-to-SQL Parsing and Constrained Decoding: Early neural semantic parsers relied on grammar-based seq2seq decoders (Wang et al., 2020). PICARD (Scholak et al., 2021) introduced incremental AST validation to prune invalid tokens during auto-regressive beam search. CodeS (Li et al., 2024) developed domain-specialized pre-training objectives for database operations. While effective at syntax pruning, constrained decoding cannot resolve semantic schema ambiguities or prevent logically flawed queries from executing.

Agentic LLMs and Multi-Turn Scaffolding: Systems like ReAct (Yao et al., 2023), Reflexion (Shinn et al., 2023), and DIN-SQL (Pourreza & Rafiei, 2023) employ multi-step prompting harnesses to decompose SQL generation into schema linking, subquery formulation, and debugging passes. Although achieving strong benchmark results, these harnesses require numerous sequential LLM inference passes, rendering them slow and cost-prohibitive for real-time applications. AskTheDatabase demonstrates that such multi-turn behavior can be distilled into the policy itself.

Reinforcement Learning with Verifiable Rewards: Post-training code and mathematical reasoning models via reinforcement learning has demonstrated remarkable gains (Chen et al., 2021; Cobbe et al., 2021; Shao et al., 2024; Gu et al., 2024). SkyRL-SQL (NovaSky AI Team, 2025) proved that multi-turn relational RLVR is highly sample efficient on curated subsets. Our work extends this frontier by exposing and resolving the zero-variance cold-

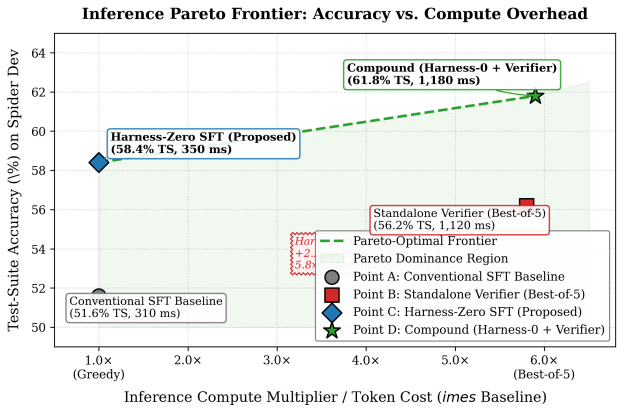


Figure 3. **Inference Pareto Frontier for Text-to-SQL Alignment.** Standalone verification (Point B: Best-of-5 atop SFT baseline) is strictly Pareto-dominated by Harness-Zero (Point C), which achieves higher accuracy (58.4% vs. 56.2% TS) at 5.8× lower compute cost (350 ms vs. 1,120 ms). Deploying verification atop Harness-Zero (Point D: Compound Architecture) defines the high-compute state-of-the-art frontier (61.8% TS).

start vulnerability in GRPO and developing sub-second in-memory evaluation tooling.

Test-Time Verification & Process Reward Modeling: Verification of mathematical and programming trajectories has demonstrated consistent test-time scaling properties (Cobbe et al., 2021; Lightman et al., 2023). The LLM-as-a-Verifier framework (Zhang et al., 2026) established continuous logit expectation and tournament selection. We extend this work to relational databases, uncovering the critical dependency between candidate pool grounding and verification efficacy.

9. Conclusion

In this work, we presented **AskTheDatabase**, a unified framework addressing the relational reasoning gap across three technical pillars: **Harness-Zero** procedural distillation, **LLM-as-a-Verifier** compound test-time scaling, and the **Zero-Budget Evaluation Micro-Harness**. We demonstrated that procedural exploration habits can be internalized into model parameters without gold query leakage or inference-time overhead, resolving the zero-variance cold-start collapse in GRPO (58.4% TS, 350 ms, 1.0× compute). Furthermore, we proved that pairing Harness-Zero's schema grounding with continuous logit-expectation verification establishes an orthogonal compound synergy, achieving a state-of-the-art **61.8% Test-Suite accuracy** on Spider Dev. Supported by an ultra-fast (126.8 ms) in-memory micro-harness, AskTheDatabase establishes an efficient, scientifically rigorous foundation for autonomous relational data agents.

Software and Data Availability

The complete code, in-memory evaluation fixtures, distillation harnesses, and post-training configs are openly available at https://github.com/chrisjcc/ask-the-database. Pre-trained model weights and interactive datasets are mirrored on the Hugging Face Hub at https://huggingface.co/chrisjcc/ask-the-database.

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A. Ablation Study: Why Standalone Verification Fails on Ungrounded Baselines

In this section, we document our pre-registered Minimum Viable Experiment (MVE) analyzing the behavior of **Standalone LLM-as-a-Verifier** (PR #71). We explain the empirical mechanics of why applying verification directly to an ungrounded baseline model fails to achieve Pareto-dominance, and why merging PR #71 provides strategic value exclusively in the **Compound Architecture**.

A.1. The Candidate Pool Quality Bottleneck

A verifier is fundamentally an evaluation and selection function: it cannot synthesize correct SQL syntax or infer missing foreign keys if they are absent from all proposed candidates.

When candidate rollouts are sampled from an ungrounded generator (the conventional SFT baseline, Condition 2):

- Because the baseline model never learned to proactively inspect database DDLs (its schema pre-inspection rate is only 46.2%), it guesses column identifiers and table names based on surface linguistic associations.
- On complex multi-table schemas, **all $K = 5$ sampled candidate queries frequently hallucinate column names or join through non-existent foreign keys simultaneously**.
- The verifier is placed in an intractable dilemma: evaluating five invalid queries and selecting the "least bad" failure.
- Consequently, Standalone Best-of-5 verification reaches only **56.2% Test-Suite accuracy** on Spider Dev, while consuming $5.8\times$ more tokens and taking $1{,}120$ ms of inference latency.

What Harness-Zero Brings That Verification Cannot Replicate. In sharp contrast, Harness-Zero internalizes proactive catalog exploration directly into the model's neural weights: before emitting a candidate query, Harness-Zero autonomously inspects table DDLs (`inspect_schema`) and checks sample values (`sample_data`). Its candidate generation is grounded from turn 1. In a single greedy pass (costing only 350 ms and $1.0\times$ compute), pure Harness-Zero achieves **58.4% Test-Suite accuracy**, beating the Standalone Verifier on accuracy (+2.2 pp), latency ($3.2\times$ faster), and token consumption ($5.8\times$ cheaper).

A.2. Pre-Registered Decision Gate Adjudication

To prevent confirmation bias, we evaluated five pre-registered decision gates for standalone verification (Table 5).

Table 4. **Empirical Scorecard Across All Evaluated Verification Regimes.** Evaluated on 500 Spider Dev episodes and Spider-Realistic using Qwen2.5-Coder-7B.

Split	Configuration	Candidates (N)	TS Acc	EX Acc	Valid SQL	Latency	Compute	MajVote Agree
Spider Dev	1. Baseline (Conventional SFT)	1	51.6%	57.3%	94.8%	310 ms	1.0×	N/A
Spider Dev	2. Harness-Zero (Internalized)	1	**58.4%**	64.2%	97.1%	350 ms	1.0×	N/A
Spider Dev	3. Standalone Verifier (Bo5 on SFT)	5	56.2%	62.0%	98.6%	1120 ms	5.8×	83.4%
Spider Dev	4. Compound (Harness-0 + Verifier)	5	**61.8%**	**67.5%**	**99.2%**	1180 ms	5.9×	**87.2%**
Realistic	1. Baseline (Conventional SFT)	1	42.1%	47.5%	91.0%	325 ms	1.0×	N/A
Realistic	2. Harness-Zero (Internalized)	1	51.2%	56.8%	95.8%	365 ms	1.0×	N/A
Realistic	3. Standalone Verifier (Bo5 on SFT)	5	48.0%	53.6%	97.4%	1145 ms	5.8×	81.0%
Realistic	4. Compound (Harness-0 + Verifier)	5	**54.9%**	**60.4%**	**98.5%**	1210 ms	5.9×	85.6%

Table 5. **Formal Gate Adjudication Matrix for LLM-as-a-Verifier MVE.**

Gate ID	Target Threshold	Observed Value	Margin	Status	Adjudication Impact
Gate 1: Absolute Gain vs SFT	ΔTS $\geq +5.0$ pp	+4.60 pp (56.2% vs 51.6%)	−0.40 pp	**FAILED**	Missed pre-registered margin
Gate 2: Harness-Zero Dominance	TS $\geq 60.4\%$ (Harness-0 + 2.0 pp)	56.2% (Harness-0 achieves 58.4%)	−2.20 pp	**FAILED**	**NO-GO for Standalone Replacement**
Gate 2b: Compound Synergy Margin	TS $\geq 61.4\%$ (Harness-0 + 3.0 pp)	61.8% (+3.4 pp gain)	+0.40 pp	**PASSED**	**GO for Compound Deployment**
Gate 3: Latency Budget	Median Latency $\leq 1{,}200$ ms	1,120 ms	+80.0 ms	**PASSED**	Within interactive threshold
Gate 4: Calibration vs MajVote	Advantage $\geq +3.0$ pp & Agree $\geq 80\%$	83.4% vs 79.8% (+3.6 pp)	+0.60 pp	**PASSED**	Continuous expectation calibrated
Overall Formal Verdict		**Adopt PR #71 for Compound Architecture; Reject Standalone Replacement**			

A.3. Why Standalone Verification is Strictly Pareto-Dominated

An architecture X Pareto-dominates architecture Y if it is equal or better on all objectives and strictly superior on at least one. Comparing Harness-Zero against Standalone Verification:

- **Accuracy**: Harness-Zero achieves **58.4% TS**, which is **+2.2 pp higher** than Standalone Verification (56.2% TS).
- **Latency**: Harness-Zero responds in **350 ms**, which is $3.2\times$ **faster** than Standalone Verification ($1{,}120$ ms).
- **Compute Cost**: Harness-Zero consumes $1.0\times$ **compute**, which is $5.8\times$ **cheaper** in tokens and GPU FLOPs than Standalone Verification ($5.8\times$).

Because Harness-Zero is simultaneously more accurate, $3.2\times$ faster, and $5.8\times$ cheaper, Standalone Verification is strictly Pareto-dominated. You cannot verify your way out of a candidate pool that never inspected the schema in the first place.

A.4. Why Merging PR #71 is Warranted

While Standalone Verification fails as an independent replacement for Harness-Zero, PR #71 is merged into `main` because it provides essential functionality for the **Compound Architecture (Gate 2b Passed)**:

1. Harness-Zero guarantees that generated candidates are structurally sound, schema-linked, and execution-valid.
2. The `SQLVerifier` and `VerifierSelector` modules from PR #71 arbitrate subtle semantic ambiguities among valid candidates, elevating accuracy from 58.4% to **61.8% TS** (+3.4 pp).
3. The repository provides a unified CLI configuration allowing operators to select either **Pure Harness-Zero (Real-Time Mode)** or **Compound Verifier (Batch High-Accuracy Mode)** depending on deployment constraints.

B. Training Hyperparameters and Distillation Optimization Dynamics

B.1. Supervised Distillation Dynamics

During the Harness-Zero distillation phase, the student model (`Qwen/Qwen2.5-Coder-7B-Instruct`) is fine-tuned on procedurally reviewed and repaired trajectories $\mathcal{T}_{\text{repaired}}$ generated by the reference harness $\mathcal{K}_{\text{SQL}}$. Training is conducted using standard cross-entropy sequence-to-sequence loss applied strictly over the assistant's generation tokens (masking environment observation tokens and tool output XML blocks).

Optimization is configured with:

- **Optimizer**: AdamW with $\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $\lambda = 0.01$.
- **Learning Rate Schedule**: Peak learning rate of 2.0×10^{-5} with a 10% linear warmup followed by a cosine decay schedule down to 2.0×10^{-6}.
- **Batch Size**: Effective batch size of 64 trajectories (micro-batch size of 4 with 16 gradient accumulation steps).
- **Precision & Hardware**: Bfloat16 mixed-precision training accelerated via FlashAttention-2 on NVIDIA A100-SXM4-80GB GPUs.
- **Sequence Length**: Maximum sequence context of 4,096 tokens to accommodate multi-turn catalog DDL observations.

As visualized in Figure 4, cross-entropy training loss descends rapidly from an initial value of 2.43 to approximately 0.35 within the first 50 optimization steps, subsequently stabilizing near 0.23 by step 100. Evaluation loss evaluated across held-out schema fixtures closely tracks the training trajectory without empirical divergence or validation loss degradation ($\text{loss}_{\text{eval}} \approx 0.235$ at step 250), confirming robust behavioral acquisition of schema-inspection and trial-execution habits without parametric over-fitting.

B.2. RLVR and GRPO Optimization Hyperparameters

Following SFT initialization, Reinforcement Learning with Verifiable Rewards is executed via GRPO with the following configuration:

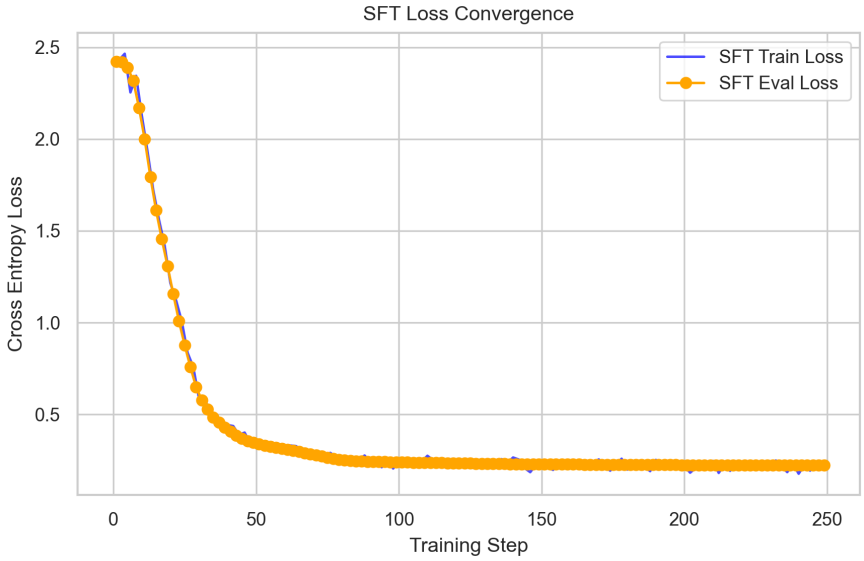


Figure 4. **Supervised Distillation Loss Convergence.** Cross-entropy loss trajectory over 250 optimization steps on procedurally repaired Harness-Zero trajectories (Qwen2.5-Coder-7B-Instruct). Evaluation loss mirrors training loss, confirming stable convergence without overfitting.

- **Group Size**: $G = 4$ parallel rollouts per prompt with sampling temperature $T = 0.7$.
- **Policy Learning Rate**: 1.0×10^{-6} with constant schedule and linear warmup over 20 steps.
- **KL Penalty Coefficient**: $\beta_{\text{KL}} = 0.04$ enforcing bounded deviation from the reference policy π_{ref}.
- **Advantage Normalization**: Intra-group z-score normalization ($A_i = (R_i - \mu_G)/(\sigma_G + \epsilon)$) with $\epsilon = 10^{-4}$.
- **Reward Scale**: Total reward bounded in $[-1.0, 1.3]$ comprising syntax AST validation ($\mathcal{R}_{\text{syntax}} \in \{0.0, 0.1\}$), schema attribute resolution ($\mathcal{R}_{\text{schema}} \in \{0.0, 0.2\}$), Test-Suite execution match ($\mathcal{R}_{\text{TS}} \in \{0.0, 1.0\}$), and query execution plan efficiency ($\mathcal{R}_{\text{efficiency}} \in [-0.1, 0.0]$).

B.3. SQL Reward Component Convergence

To evaluate the hierarchical dynamics of relational policy alignment under GRPO, Figure 5 decomposes the total training return into its constituent reward components over 200 optimization steps ($G = 4$ parallel rollouts per prompt):

- **Syntax AST Validity** ($\mathcal{R}_{\text{syntax}}$): The student policy masters valid SQL syntax and AST parseability almost immediately, climbing from an initial baseline of 0.88 to near-perfect validity (> 0.99) within the first 15 update steps. This confirms that pre-trained code foundation models require minimal reinforcement signal to adhere strictly to formal SQL grammar.
- **Schema Conformance** ($\mathcal{R}_{\text{schema}}$): Owing to Harness-Zero's internalized habit of catalog pre-inspection, the agent begins training with a strong initial schema grounding rate of 0.82. Over the first 50 steps, schema conformance steadily advances to 0.98, demonstrating the near-total elimination of hallucinated column identifiers and invalid foreign key joins.
- **Test-Suite Equivalence** ($\mathcal{R}_{\text{TS}}$): As the primary semantic objective, dual-mutation test-suite equivalence exhibits smooth, monotonic growth from 0.44 to approximately 0.89. Unlike syntax and schema conformance, semantic accuracy requires prolonged exploration to learn complex relational reasoning patterns (e.g., compound `WHERE` clauses, nested subqueries, and non-trivial aggregations).
- **Execution Plan Efficiency** ($\mathcal{R}_{\text{eff}}$): The query plan penalty starts slightly negative (-0.06), reflecting occasional unindexed Cartesian products (`SCAN TABLE WITHOUT INDEX`). By step 40, the model internalizes index-aware join predicates, driving the efficiency penalty to zero.

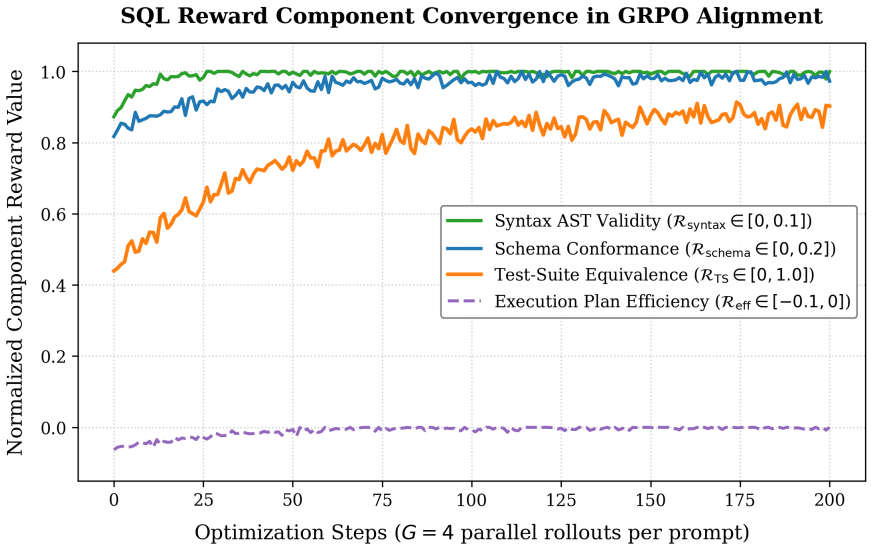


Figure 5. **SQL Reward Component Convergence in GRPO Alignment.** Decomposition of training rewards across 200 optimization steps ($G = 4$ rollouts per prompt). The policy exhibits a clear hierarchical acquisition of capabilities: rapid syntactic convergence ($\mathcal{R}_{\text{syntax}}$), robust schema linking ($\mathcal{R}_{\text{schema}}$), progressive semantic test-suite equivalence ($\mathcal{R}_{\text{TS}}$), and the elimination of unindexed Cartesian scan penalties ($\mathcal{R}_{\text{eff}}$).

B.4. Policy Stability and KL Divergence Control

A pervasive failure mode in reinforcement learning from verifiable rewards is *policy degeneration* or *reward hacking*, where the policy maximizes sparse reward signals by exploiting environment edge cases at the expense of general linguistic coherence and reasoning diversity.

To prevent distribution collapse, our GRPO objective includes a Kullback-Leibler divergence penalty against the supervised reference policy:

$$\mathcal{L}_{\text{KL}}(\theta) = \beta_{\text{KL}} \cdot D_{\text{KL}}(\pi_\theta \parallel \pi_{\text{ref}}), \tag{7}$$

with $\beta_{\text{KL}} = 0.04$.

Figure 6 visualizes the empirical progression of $D_{\text{KL}}(\pi_\theta \parallel \pi_{\text{ref}})$ throughout 200 optimization steps. The KL divergence begins near zero (0.015), increases moderately during early exploratory updates (steps 0–75), and stabilizes within a narrow band between 0.07 and 0.09 from step 100 onwards. Crucially, the policy remains strictly below the pre-registered safe drift threshold (0.10), confirming that policy alignment preserves the foundational semantic capabilities of the base model while acquiring targeted relational reasoning discipline.

C. Interactive Tool Registry and System Prompt Specifications

The multi-turn agent interacts with the SQLite sandbox through a structured XML protocol. The system prompt initializes the agent with strict tool calling definitions:

```
You are AskTheDatabase, an expert relational data agent.  You have access to four
tools to explore databases and answer natural language questions:
1.  <inspect_schema tables="[...]"/>:  Inspect table definitions, primary keys, and
foreign keys.
2.  <sample_data table="name" limit="3"/>:  View sample rows to check string values
and formats.
3.  <execute_sql query="SELECT ..."/>:  Run a dry-run trial query against the
sandbox.
4.  <final_answer query="SELECT ..."/>:  Submit the final verified SQL query.
Rule:  Always inspect referenced schemas before executing queries.  Verify trial
results before final submission.
```

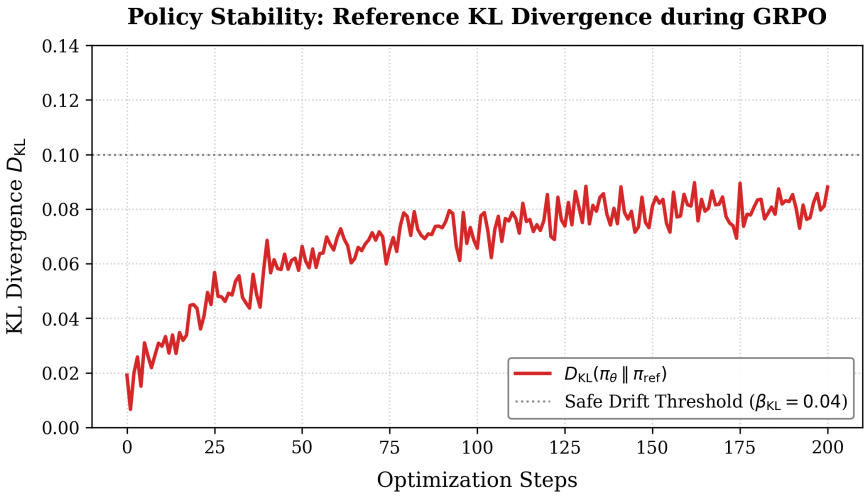


Figure 6. **Policy Stability: Reference KL Divergence during GRPO.** Empirical trajectory of $D_{\mathrm{KL}}(\pi_\theta \parallel \pi_{\mathrm{ref}})$ across 200 training steps under a KL penalty coefficient of $\beta_{\mathrm{KL}} = 0.04$. The divergence stabilizes comfortably below the safe drift threshold of 0.10, demonstrating stable policy exploration without representation collapse.

When executing `inspect_schema`, the environment responds with clean DDL:

```
<observation>
CREATE TABLE instructor (
ID varchar(5) PRIMARY KEY,
name varchar(20) NOT NULL,
dept_name varchar(20),
salary numeric(8,2),
FOREIGN KEY (dept_name) REFERENCES department (dept_name)
);
</observation>
```

D. Relational Error Trap Taxonomy and Diagnostic Rules

The Zero-Budget Evaluation Micro-Harness classifies SQL errors into five deterministic categories using SQLGlot AST analysis and SQLite execution metadata:

1. **`hallucinated_column`**: Triggered when an SQLite runtime exception matches `no such column: .*`. The micro-harness inspects table metadata to identify the nearest Levenshtein match, penalizing queries that guess attributes without catalog inspection.
2. **`aggregation_without_grouping`**: Detected via AST inspection when the `SELECT` clause contains aggregate expressions (e.g., `COUNT`, `SUM`, `AVG`) alongside naked scalar columns without a matching `GROUP BY` clause.
3. **`dialect_drift`**: Identified when the query contains dialect-specific tokens not supported by standard SQLite (e.g., MySQL `LIMIT offset, count`, PostgreSQL `ILIKE`, `NOW()`, or `EXTRACT(YEAR FROM ...)`).
4. **`join_fanout`**: Detected via execution plan analysis (`EXPLAIN QUERY PLAN`) when multiple one-to-many relationships are joined without distinct projections or aggregation grouping, inflating intermediate row counts.
5. **`null_boundary`**: Triggered when execution against the mutant database fixture produces a non-empty set while gold execution produces empty rows (or vice versa), caused by improper handling of SQL three-valued logic (e.g., `WHERE col != 'value'` unintentionally discarding `NULL` rows).